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Parameters design for vertical section of a 30,000-ton heavy-haul railway aligment based on longitudinal dynamics analysis

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***Abstract*—Railways, crucial to global logistics (transporting 40% of supply chain materials), face competitive pressures from road and air freight. This necessitates technological advances in heavy haul operations. China's network, featuring 10,000-ton trains and milestone 30,000-ton tests, is constrained by outdated standards (e.g., TB 10625-2010), which limit axle loads to ⩽25 tons and traction to ⩽20,000 tons, below modern 30-ton/30,000-ton requirements.**

This study employs UM dynamic simulation to model 30-ton axle load, 30,000-ton traction trains. Guided by design codes and longitudinal dynamics theory, it systematically analyzes dynamic characteristics:

Reviewing heavy-haul literature and norms to establish combined slope parameter ranges (gradient, length, vertical curve radius) and evaluation indices (max coupler force, acceleration, braking distance).

Simulating three 30,000-ton train formations (1+1+1, 2+2, 3+0) per design specifications.Analyzing longitudinal impulse and braking distance changes during emergency braking (60/80 km/h) on single slopes for each formation.Quantifying braking distance and acceleration data variation (20-100 km/h) for the 1+1+1 formation.Investigating combined slope parameter impacts on longitudinal dynamics using the 1+1+1 model.

Keywords—Heavy-haul railway, Longitudinal dynamics, Vertical alignment, 30000-ton train, Emergency braking

I. INTRODUCTION

Railways are crucial for global trade, dominating freight transport (over 38% of global land volume) and vital for resource-based economies. China's heavy-haul systems, notably Daqin Railway, achieved breakthrough 30,000-ton train operations in 2014 (Yan, 2016). However, these ultra-long trains (>2 km) intensify technical challenges: braking response time increases 1.8x and coupler force fluctuations rise 40% on complex gradients (Knutton, 2004), demanding optimization of longitudinal dynamics and braking coordination. This technology boosts single-train capacity 300% versus 10,000-ton trains, significantly enhancing efficiency (e.g., 0.28 tons standard coal/10,000 ton-km vs. 1.05 tons for road transport).

Therefore, this research focuses on 30-ton axle load, 30,000-ton traction trains. Using longitudinal dynamics theory and UM simulation software, it models the influence of slope parameters and train formation on dynamic performance.

The innovation of this research lies in the utilization of UM and longitudinal dynamics to analyze the performance of train formation modes through the study of braking distance and maximum acceleration; and a thorough investigation of the relationship between braking distance and initial speed was conducted by drawing graphs.

II. PARAMETER ANALYSIS ANDEVALUATION INDICATORS

Existing studies indicate that excessive acceleration in heavy-haul trains may induce derailment risks, while extreme coupler forces could lead to structural failures. Consequently, peak coupler forces and longitudinal acceleration have become critical evaluation metrics for train longitudinal dynamics in engineering practice. Investigating braking distance further enhances operational safety by determining required safe braking zone lengths during deceleration. This study therefore selects coupler force, acceleration, and braking distance as primary evaluation parameters.

The China Academy of Railway Sciences has established operational thresholds for maximum coupler forces and accelerations in large-axle-load railways, as summarized in Table I below.

TABLE I. RESEARCH EVALUATION CRITERIA

Operating conditions	***Maximum hook force/kN***	***Longitudinal acceleration***
Normal operating conditions	≤1000	<9.8 m/s2
Common braking conditions	≤1500	
Emergency braking conditions	≤2250	

This chapter initiates with verification of heavy-haul train traction capacity, followed by static theoretical analysis of individual slope parameters through literature review: gradient, grade difference, vertical curve radius, and grade segment length. Under defined assumptions, the research scope for these parameters was established alongside development of a rational evaluation index system, thereby laying the methodological foundation for subsequent investigations.

TABLE II. RANGE OF TRACK PROFILE PARAMETERS

Longitudinal section parameters of the line	Range
Single slope gradient	≤9‰
Slope difference	≤18‰
Radius of the vertical curve	≤40000m
Slope section length	≤4000m

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III. ESTABLISHMENT OF 30,000-TON HEAVY HAUL TRAIN MODEL

1)Train dynamics theory:Compared to single-locomotive dynamics, train dynamics builds upon unit vehicle models to examine longitudinal mechanical coupling in multi-vehicle formations. As carriages may occupy distinct track segments with asynchronous traction/braking force application timing, the research focus centers on modeling and optimizing longitudinal dynamic responses under multiple operational scenarios. The following presents force equilibrium analysis for an isolated vehicle unit:

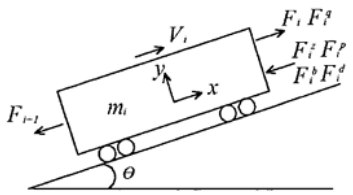


Fig.1. Single Car Force Model

$$m_i^* x_i = F_i^{\mathrm{r}} + F_i^q + F_i^q - F_{i+1}^z - F_i^p - F_i^b - F_i^d \quad (1)$$
$$(i = 1,2,\ldots n)$$
$$F_i^z = F_i^z(x_1, x_1) \quad (2)$$
$$F_i^p = F_i^p(x_1) \quad (3)$$
$$F_i^q = F_i^q(x_1) \quad (4)$$
$$F_i^b = F_i^b(x_1, t) \quad (5)$$
$$F_i^d = F_i^d(x_1, t) \quad (6)$$

mi=The mass of locomotives and freight cars (t)
F_i^{r}=Longitudinal coupler force of the train (kN)
F_i^z=The allowable basic resistance (kN) of the locomotive
F_i^p=The force provided by the ramp (kN)
F_i^q=The traction force provided by the locomotive (kN)
F_i^b=Aerodynamic force (kN)
F_i^d=Dynamic braking force (kN)
x_1=Displacement of locomotives and freight cars (m)
t=Braking time (s)

2)Development of longitudinal dynamics simulation model:Prior to UM simulation analysis, key modeling assumptions are defined:

Geometrically accurate vehicle dimensions

Negligible coupler/buffer mass

Center of mass at geometric center

Wheel-rail friction coefficient:0.25;external forces applied at mass center

Simulations employ the FXD2B electric locomotive, engineered for 30-ton axle load heavy-haul operations with multi-locomotive traction capability. Technical specifications include: length 18.975m, 8-axle configuration, 9,600 kW traction power, and 100 km/h maximum operating speed.

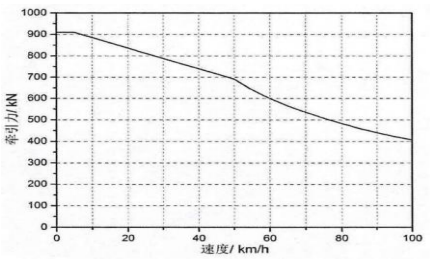


Fig.2. FXD2B Electric Locomotive Traction Characteristics

China's primary freight models have evolved from C80/C96 to higher-capacity designs. This study analyzes 30,000-ton train dynamics using C100 freight cars with key specifications: 25t axle load; 12.5m length; 240-unit formation (3,000m total).

UM simulations model braking wave propagation at 280 m/s (service) and 300 m/s (emergency).

3)Braking force and Train running resistance assumptions：The FXD2B locomotive and C100 freight cars utilize standard pneumatic braking, employing compressed air for brake control and synchronization via pressure propagation. Braking wave velocities are:

Service braking release: 280 m/s

Emergency braking: 300 m/s

The braking force for individual vehicles is calculated as:

$$F = \mu \cdot P \cdot A$$

F=Braking force of a single vehicle (kN)
μ=The friction coefficient of the brake shoe (generally 0.25-0.35, taken as 0.3 here)
P=Brake lever pressure (3.5bar for common brakes)
A=Effective area of the brake bar (assuming 0.016m ²)

$$F = \mu \cdot P \cdot A$$

F =0.3×350000×0.016=1680N

The entire vehicle has a total of 240 carriages. The initial estimated total braking force is: 240×1680=403 kN

Train running resistance refers to the sum total of forces opposing motion during operation, which dissipates train kinetic energy and requires compensation through locomotive traction or braking forces. Resistance comprises two categories: basic and additional resistance. This study disregards additional resistance, focusing exclusively on basic resistance components.

Basic resistance consists of:

1) Starting resistance
2) Running basic resistance
Per the Train <<Traction Calculation Regulations>>, reference values for unit starting resistance are:
- FXD2B electric locomotive: ω_q'=5 N/kN
- C100 freight car:ω_q''= 6 N/kN
According to the definition of the basic operating resistance in the "Train Traction Calculation Code" :
ω_0'Basic resistance of locomotive operation，$\omega_0' = 0.85 + 0.0120v + 0.000300v^2$(N/kN)，$v$ is velocity.

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w_0'' is the basic resistance for the operation of a truck unit, which is related to the vehicle type and speed, w_0''=0.90 + $0.050v + 0.000130v^2$(N/kN), v is velocity.

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4)Train formation configuration : Heavy-haul train formation design structures locomotive/freight car arrangements to enhance high-tonnage efficiency. It coordinates locomotive placement (head/distributed), braking synchronization, and sequencing to reduce longitudinal forces and braking delays while balancing loading rates and track capacity. Critical safety considerations include coupler-buffer robustness and thermal load management under complex conditions.

To investigate formation effects on longitudinal impacts - particularly braking distance optimization - three configurations are analyzed:

1+1+1 : FXD2B + 120×C100 + FXD2B + 120×C100 + FXD2B

3+0 formation: 3×FXD2B + 240×C100

2+2 formation: 2×FXD2B + 120×C100 + 2×FXD2B + 120×C100.

To verify whether my grouping method is feasible, I select my 1+1+1 grouping: FXD2B+120 x C100+FXD2B+120 x C100+FXD2B. Emergency braking was carried out at an initial speed of 60km/h on a straight road, referring to the simulation results of operation under the same conditions in relevant literature. The grouping method of the relevant literature is: HXD1+150 x C80+HXD1+150 x C80+HXD1

The comparison data of the simulation results in this paper with those in the literature are shown in Table III as follows:

TABLE III. COMPARISON RESULTS

	Maximum hook force	Maximum acceleration	Braking distance
Literature simulation results	581kN	3.77m/s^2	609m
The simulation results of this paper	462kN	2.91m/s^2	378m

IV. LONGITUDINAL IMPULSE ANALYSIS OF TRAINS ON A SINGLE SLOPE

To compare formation modes' impact on longitudinal dynamics and braking safety, this study first analyzes train impulse on single slopes under emergency braking (60 km/h, 300 m/s wave velocity). Three formations are evaluated: 1+1+1, 2+2, and 3+0.

1)*Ascending grade:*Since the slope value of a single ramp section cannot exceed 9 per thousand and the length of a single ramp section cannot exceed 4000 meters, for the setting of the uphill ramp, the slope is 4.5‰ and the length of the ramp section is 3200 meters.

Three traction modes were respectively operated on the uphill track. Their maximum hook force, maximum acceleration and braking distance were recorded. The specific data are shown in Table IV below:

TABLE IV. DATA COMPARISON ON 4.5‰ ASCENDING GRADE

Formation	Maximum hook force	Maximum acceleration	Braking distance
1+1+1	466kN	2.971m/s^2	349m
3+0	1099kN	5.8m/s^2	375m
2+2	923kN	3.83m/s^2	359m

Table IV data confirm all values remain within prescribed safety thresholds. During uphill emergency braking, the 1+1+1 formation exhibits minimal longitudinal impulse and shortest required braking slope, while 3+0 shows maximal impulse and longest slope. Braking distance comparisons (Fig.3.) were generated via UM data exported to Excel andprocessed with Python.

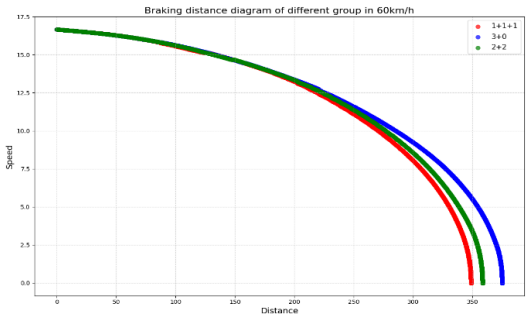


Fig.3. Braking Distance Comparison on 4.5‰ Ascending Grade

It can be intuitively seen from the figure that the braking distance of the 1+1+1 formation is the shortest, which is only 349m. The 3+0 formation mode has the longest braking distance, which is 375m.

2)Level track:The length of the slope section of the level track is 3,200 meters.

Run in three traction modes respectively on a straight track. Record their maximum hook force, maximum acceleration, and braking distance. The specific data are shown in Table V below:

TABLE V. DATA COMPARISON ON LEVEL TRACK

Formation	Maximum hook force	Maximum acceleration	Braking distance
1+1+1	462kN	2.91m/s^2	378m
3+0	1072kN	5.92m/s^2	410m
2+2	913kN	3.85m/s^2	387m

Table V confirms all values comply with safety thresholds. On straight tracks, the 1+1+1 formation maintains minimal longitudinal impulse and shortest braking slope, while 3+0 exhibits maximal impulse and longest slope. Braking distance comparisons are shown in Fig.4.(generated via UM data exported to Excel and processed with Python).

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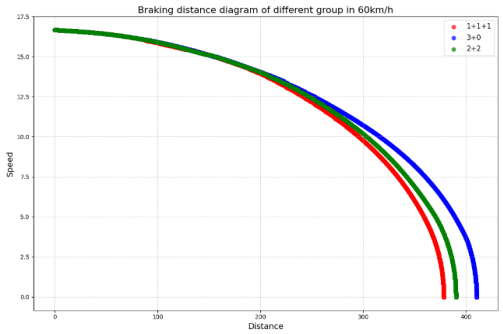


Fig.4. Braking Distance Comparison on Level track

It can be intuitively seen from the figure that the 1+1+1 grouping method has the shortest braking distance, which is only 378m. The 3+0 formation mode has the longest braking distance, which is 410m.

3)Descending grade:Since the slope value of a single ramp section cannot exceed 9 per thousand and the length of a single ramp section cannot exceed 4000 meters, for the setting of the downhill ramp, the slope is 4.5‰ and the length of the ramp section is 3200 meters.

Run in three traction modes respectively on the downhill ramp. Record their maximum hook force, maximum acceleration, and braking distance. The specific data are shown in Table VI below:

TABLE VI. DATA COMPARISON ON 4.5‰ DESCENDING GRADE

Formation	Maximum hook force	Maximum acceleration	Braking distance
1+1+1	460kN	2.89m/s^2	412m
3+0	1055kN	6.11m/s^2	452m
2+2	905kN	3.88m/s^2	428m

Table VI confirms compliance with all safety thresholds. During downhill emergency braking, the 1+1+1 formation demonstrates minimal longitudinal impulse and shortest braking slope, while 3+0 requires maximal impulse and longest slope. Comparative braking distances are shown in Fig.5.

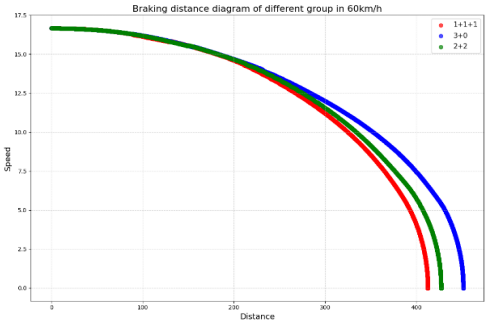


Fig.5. Braking Distance Comparison on 4.5‰ Descending Grade

It can be intuitively seen from the figure that the 1+1+1 grouping method has the shortest braking distance, which is only 412 meters. The 3+0 formation mode has the longest braking distance, which is 452 meters.

Conclusion:Emergency braking tests at 60 km/h on single slopes confirmed all three formations remain within safety thresholds for coupler force and acceleration, establishing operational viability. Across all gradients, the 1+1+1 configuration demonstrated minimal longitudinal impulse and required the shortest braking slope.

V. TRENDS OF INITIAL VELOCITY AND BRAKING DISTANCE CHANGES

This study analyzes FXD2B heavy-haul train braking dynamics using a fixed 1+1+1 formation on descending grades. Braking distances were systematically measured across nine initial speeds (20-100 km/h). Python-based nonlinear regression derived a statistically significant speed-distance relationship, providing a basis for safe braking distance design and grade length standardization. Data in Table VII and Fig.6.

TABLE VII. BRAKING DISTANCE VARIATION WITH INITIAL SPEED ON DOWNWARD GRADE

Initial Speed km/h	Barking distance m
20	69.18
30	130.8
40	208.3
50	302.4
60	412.58
70	538.81
80	681.18
90	840.3
100	1016.4

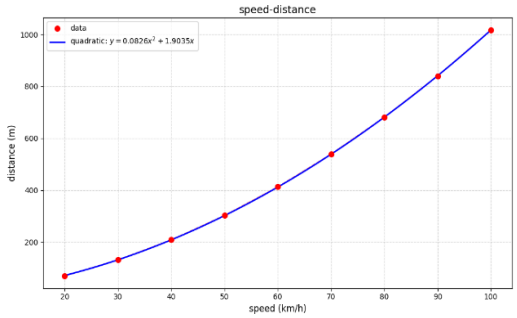


Fig.6. Downward Braking Trend with Initial Speed Variation

This study employs controlled-variable braking dynamics analysis across track profiles. Curve fitting of single-grade descent data revealed a quadratic braking distance-velocity relationship. Comparative experiments on level and ascending grades validated this function's universality.

TABLE VIII. BRAKING DISTANCE VARIATION WITH INITIAL SPEED

Initial Speed km/h	Downhill barking distance m	Uphill barking distance m	Level track barking distance m
20	69.18	55.75	61.9

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30	130.8	108.145	118.632
40	208.3	174.4	190.05
50	302.4	254.3	276.44
60	412.58	348.56	378.24
70	538.81	456.6	494.7
80	681.18	578.5	626.06
90	840.3	714	772.4
100	1016.4	863.6	933.9

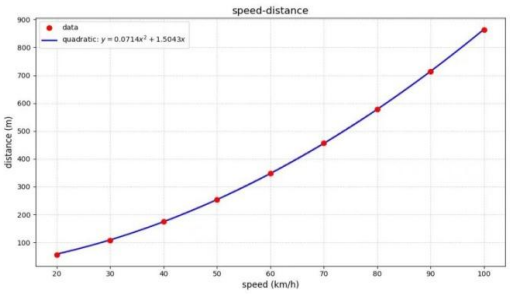


Fig.7. Upward Braking Trend with Initial Speed Variation

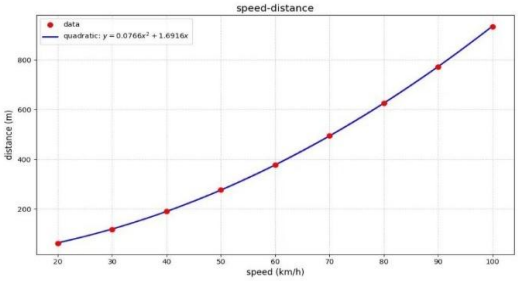


Fig.8. Level Track Braking Trend with Initial Speed Variation

Under constant parameters (formation mass, braking levels), systematic braking distance-velocity data confirm a universal quadratic relationship across gradients. Validation tests on ascending and level grades demonstrate parabolic growth of braking distance with initial velocity, providing theoretical basis for safety distance calculations.

VI. Trends of Initial Velocity and Maximum Acceleration Variation

This study investigates train dynamic response characteristics under single-grade conditions, building upon the established quadratic relationship between braking distance and initial velocity to further analyze dynamic variations of maximum acceleration with speed changes. Using the controlled variable method with fixed parameters (formation mass, grade gradient), real-time acceleration data were systematically collected across discrete initial speed intervals (20-100 km/h). Results establish functional relationships between acceleration extremes and velocity changes, supporting optimized grade control strategies and speed-dependent safety thresholds. These findings substantially enhance heavy-haul railway operational safety and energy efficiency.

TABLE IX. Maximum Acceleration Variation with Initial Speed

Initial Speed km/h	Downhill max acceleration m/s^2	Uphill max acceleration m/s^2	Level track max acceleration m/s^2
20	2.76	2.77	2.79
30	2.43	2.76	2.78
40	2.413	2.73	2.73
50	2.62	2.7	2.7
60	2.603	2.62	2.66
70	2.57	2.59	2.65
80	2.587	2.55	2.59
90	2.47	2.47	2.57
100	2.24	2.45	2.55

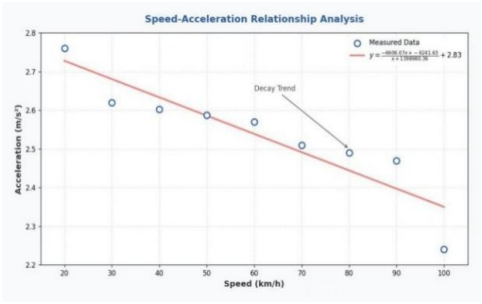


Fig.9. Downward Maximum Acceleration Trend with Initial Speed Variation

Fig.9. demonstrates that the maximum acceleration of heavy-haul trains on single-grade sections exhibits limited sensitivity to speed variations, though a general inverse correlation is observed - acceleration decreases as initial speed increases. To validate this relationship's universality, controlled experiments were conducted on level track and standard upward grades (4.5‰).

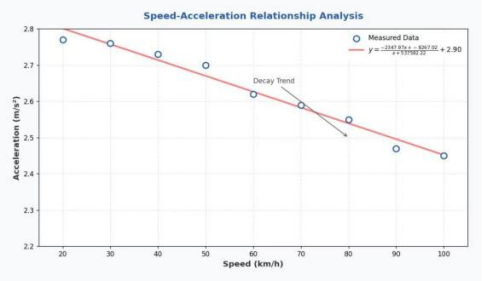


Fig.10. Upward Maximum Acceleration Trend with Initial Speed Variation

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Fig.11. Level Track Maximum Acceleration Trend with Initial Speed Variation

VII. CONCLUSIONS

This study analyzes heavy-haul train dynamics under varying gradients, evaluating formation configurations, speeds, and slopes on braking distance, acceleration, and coupler forces. Key findings:

Formation Optimization: 1+1+1 configuration demonstrated minimal longitudinal impulses and shortest braking zones across gradients, with <1% coupler force variation between slope types. Downward grades yielded higher peak accelerations (-2.9 m/s²).

Braking Distance-Velocity Relationship: Braking distance exhibits quadratic growth with velocity ($S \propto v^2$), increasing 30% per 10 km/h speed increment. Acceleration intensifies >60 km/h.

Acceleration Dynamics: Maximum acceleration decreases with increasing speed (e.g., 20% reduction from 20 to 100 km/h on level track), validated across gradients, potentially linked to speed-dependent wheel-rail adhesion.

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